Procedural Istanbul

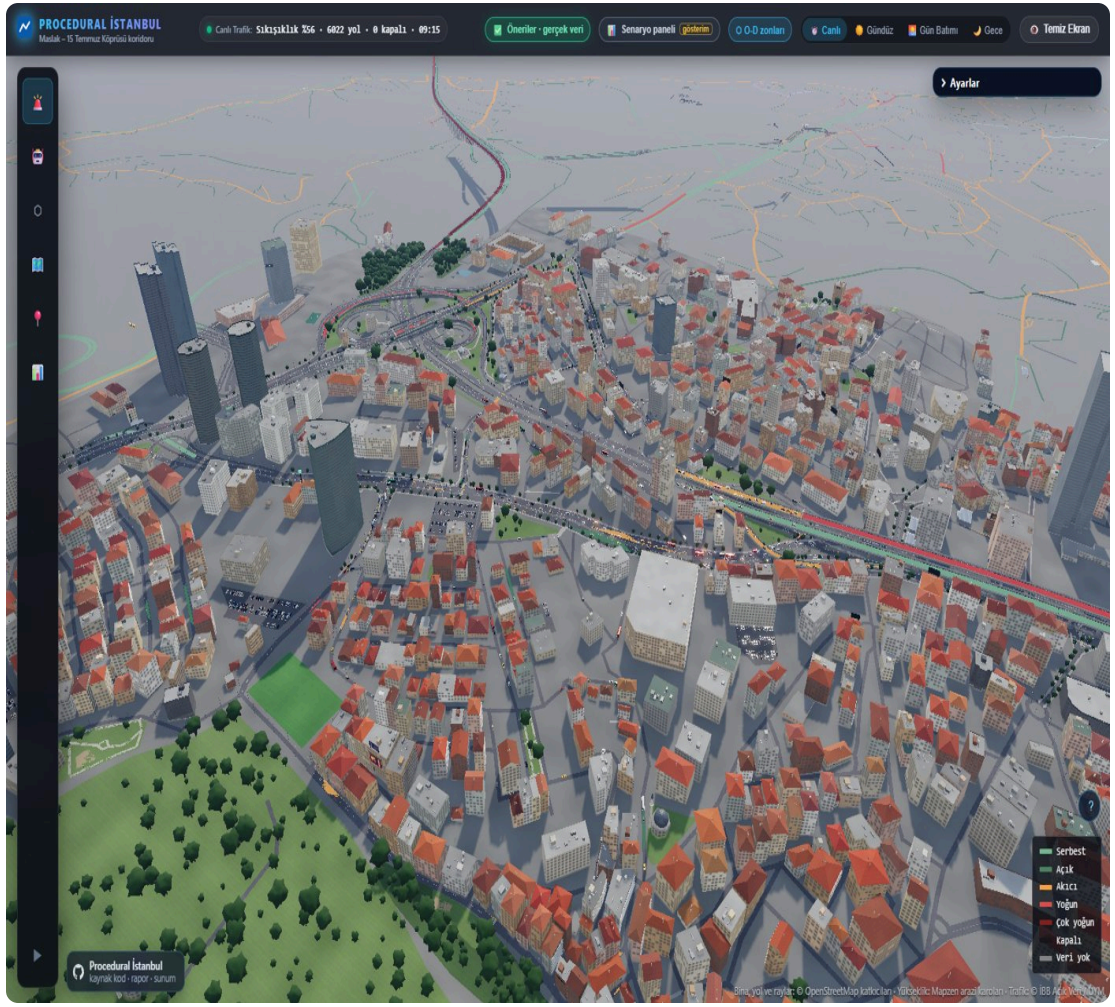
Origin–destination estimation from censored speed observations, and an open digital twin of the Maslak–15 July corridor

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Interim results · October 2026

github.com/aaasiroglu/procedural-istanbul (MIT) · 3D engine derived from Yong Su's "Tokyo" project

The corridor: Maslak – Levent – Mecidiyeköy – Beşiktaş – 15 July Bridge



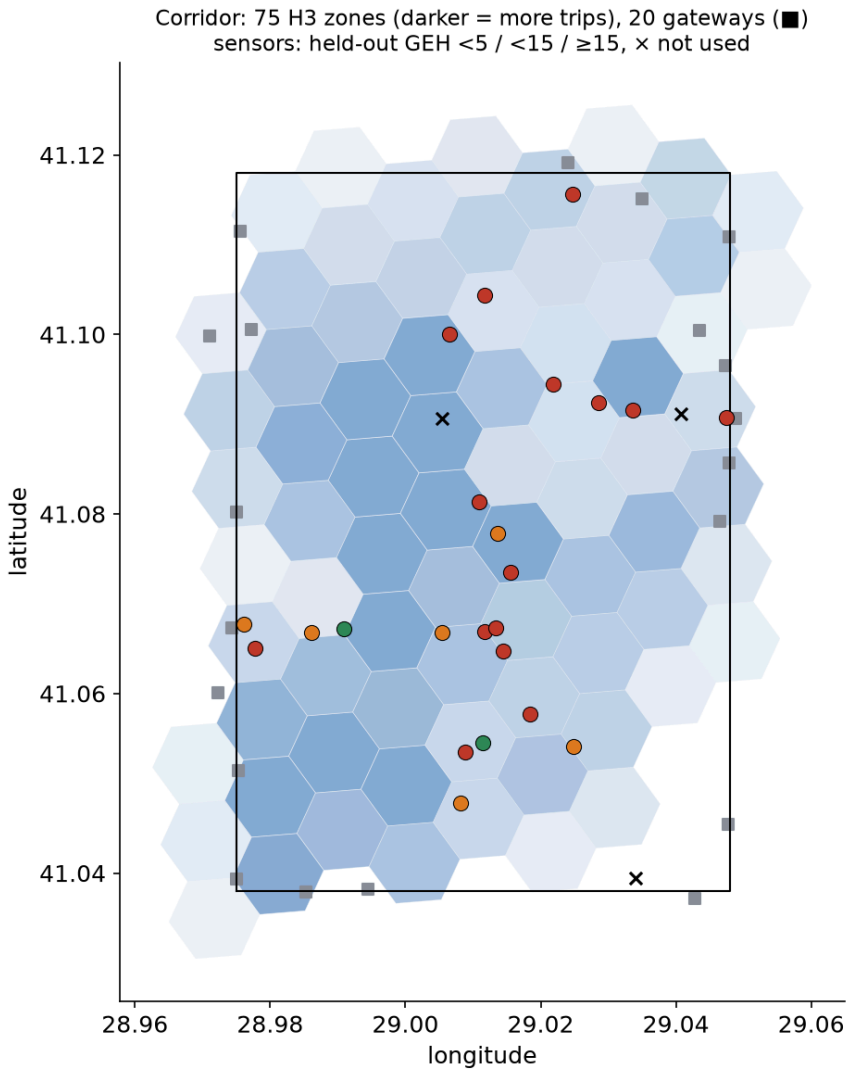
- A **3D digital twin** running in the browser: roads, buildings and terrain compiled entirely from open data (OpenStreetMap, SRTM).
- Road segments are coloured by İBB's **live congestion class** (free flow → very heavy).
- Underneath runs a **transport model**: a SUMO network, inverse OD estimation and validation against İBB counts.
- Research question: **how reliably can the demand matrix a simulator needs be inferred from a sensor network that measures only speed?**

Problem

- **A simulator needs trips; sensors deliver speeds.** A simulator such as SUMO needs an origin–destination (OD) matrix: who travels from where to where. İBB's live feed gives each road segment only a **speed** and a **congestion class**.
- **Few counts, an ill-posed problem.** There are 9,025 OD pairs, hundreds of times more than the observations. The data alone cannot determine the matrix.
- **The weakness of the common approach.** Turning speeds into pseudo-counts injects a precision the data does not carry. Most studies do not report how much of an estimate comes from the data and how much from assumptions.

> **Aim:** estimate the corridor's hourly OD matrix from open data, report its uncertainty explicitly, and validate it against counts the model has not seen.

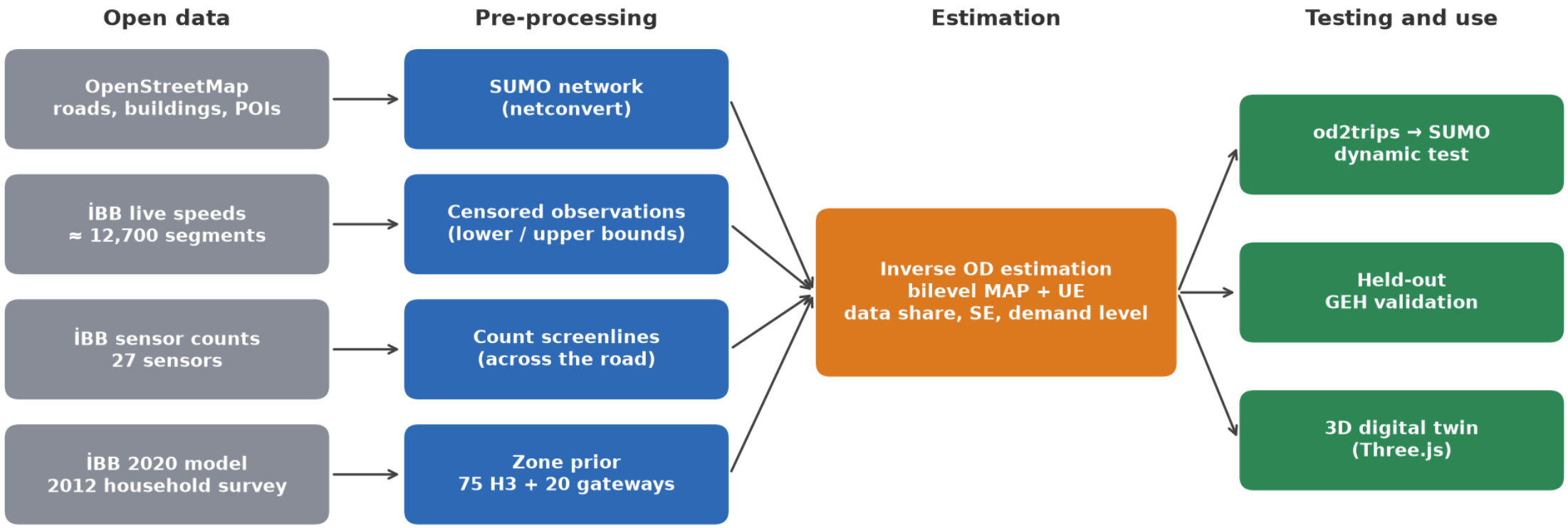
Study area and data



Area: Beşiktaş – Maslak and Çağlayan – 15 July Bridge (≈ 5.9 × 8.9 km)

Data	Source
Road network, buildings, POIs	OpenStreetMap
Live segment speeds (≈ 12,700)	İBB Traffic Management Centre
Daily sensor counts (27 sensors)	İBB Open Data
Hourly probe-vehicle density	İBB Open Data
Calibrated transport model (2020)	İBB Open Data
Household travel survey (2012)	İBB Open Data
Digital elevation model	Mapzen / SRTM

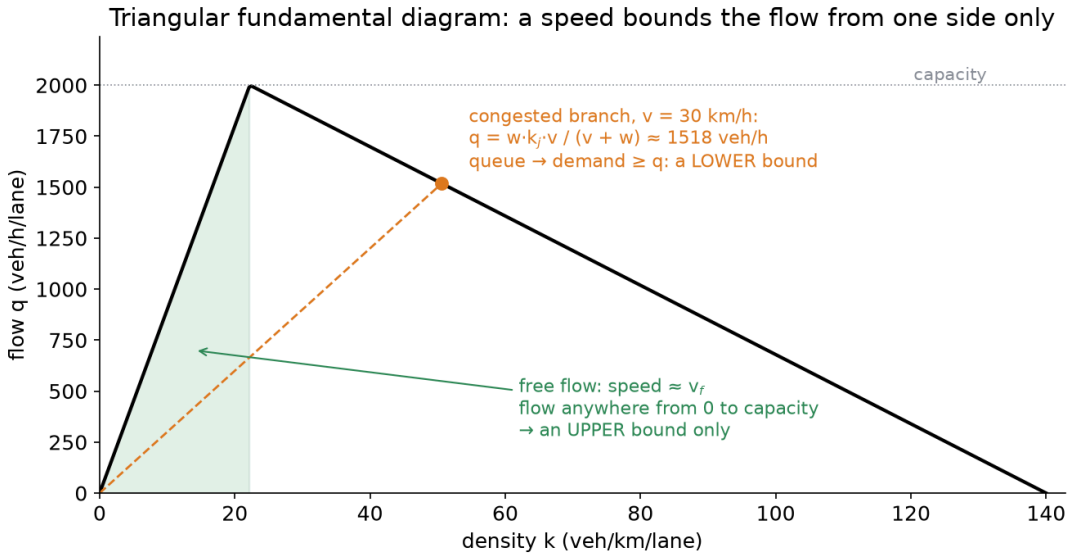
Architecture



Every step is reproducible with a single command (`npm run sumo:net` , `sumo:map-ibb` , `od:corridor`). The inverse OD method is tested by 32 synthetic checks.

1 · Method

Contribution 1: a speed is a censored observation



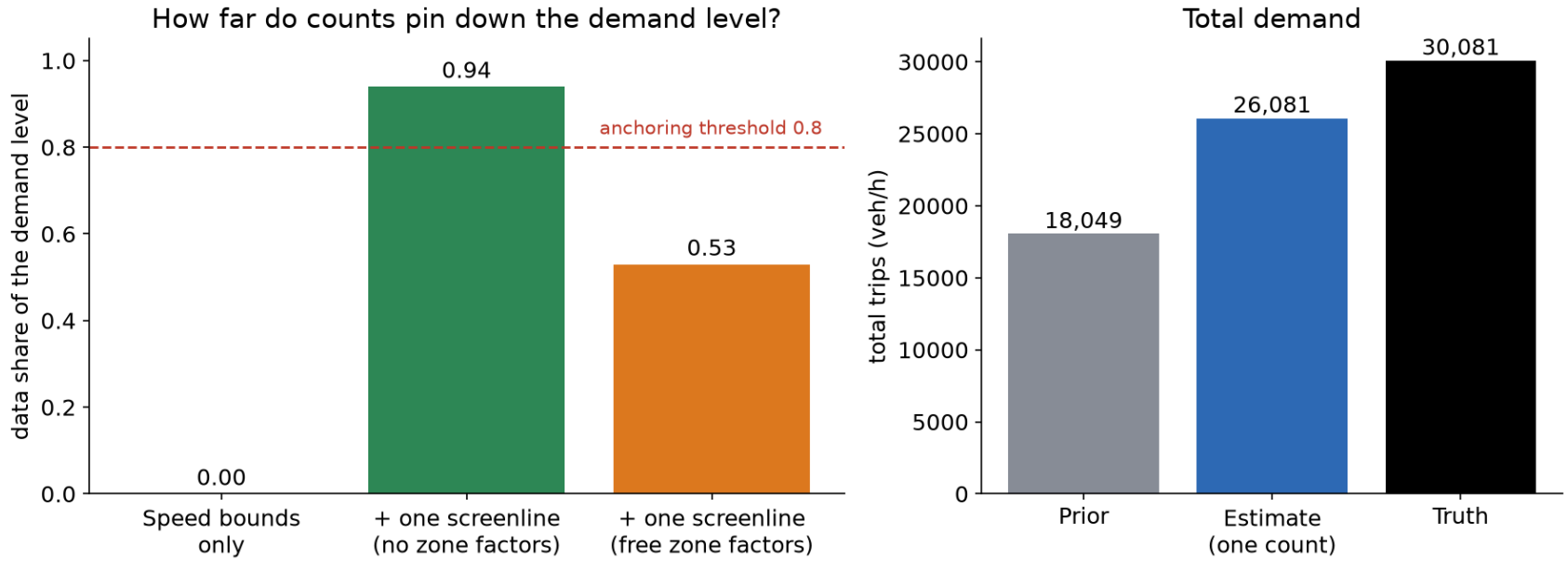
- **A free-flow speed does not determine the flow:** any flow between 0 and capacity is possible. The speed gives only an **upper bound**.
- **On the congested branch** $q = w \cdot k_j \cdot v / (v + w)$. A queue means demand exceeds the flow getting through, so the speed gives a **lower bound**.
- Whether a segment is congested is decided by İBB's own per-segment **congestion class**.

$$\min_{x \geq 0} \sum_{\text{counts}} \frac{((Px)_e - y_e)^2}{s_e^2} + \sum_{\text{lower}} \frac{\max(0, y_e - (Px)_e)^2}{s_e^2} + \sum_{\text{upper}} \frac{\max(0, (Px)_e - y_e)^2}{s_e^2} + \mathrm{KL}_\lambda(x \,\|\, x_0)$$

Contribution 2: uncertainty and honest validation

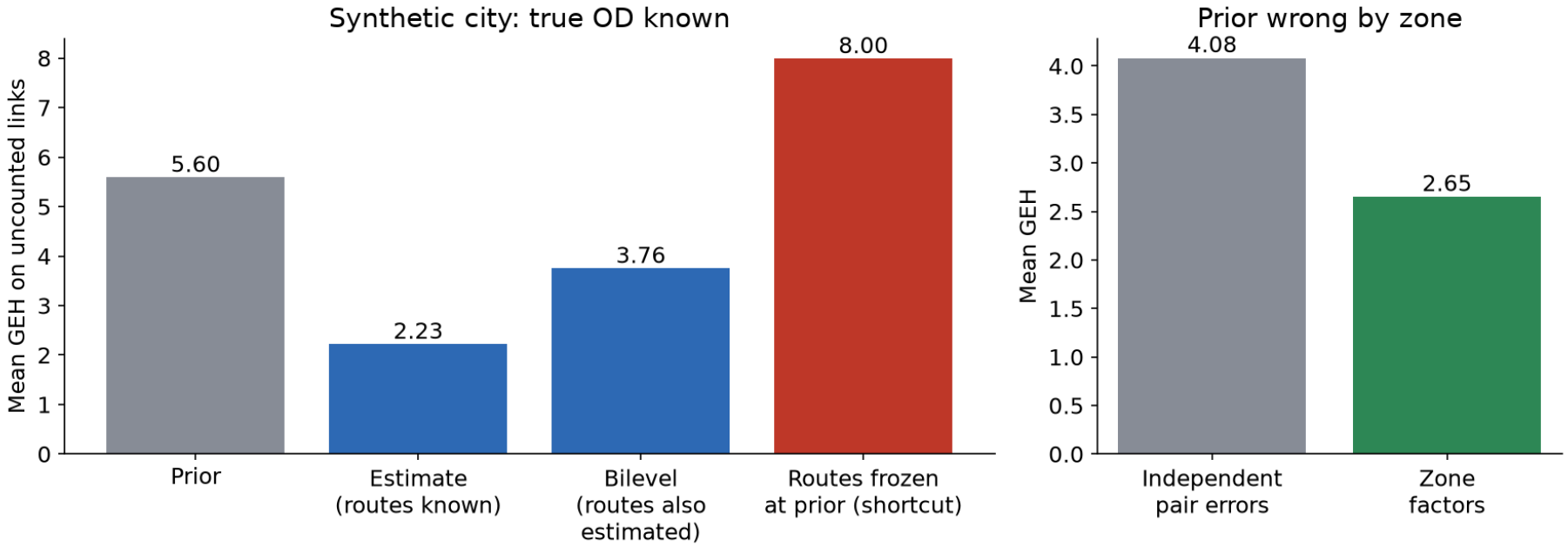
- **Data share.** For each OD pair, the part of the estimate that comes from the data: $1 - \sigma^2_{\text{posterior}} / \sigma^2_{\text{prior}}$. Laplace approximation; the Woodbury identity reduces an n × n inversion to an m × m solve.
- **Correlated prior errors.** $x_{ij} = x_{0,ij} \cdot e^{g + a_i + b_j + u_{ij}}$. Zone factors ($a_i$, b_j) correct a zone's whole traffic with a single number; **g** carries the level of the whole matrix.
- **Held-out validation.** GEH is measured not on the counts the model was fitted to, but on each count left out in turn.
- **Bilevel estimation.** UE assignment and estimation alternate; oscillation is damped by the method of successive averages (Spiess 1990; Yang et al. 1992).
- **Numerical conditioning.** Optimisation in log space ($z = x_0 \cdot e^u$); L-BFGS converges even when prior values span twelve orders of magnitude.

Contribution 3: when is the level of demand anchored?



- The data share of the level comes from the curvature of the profile objective: $\sigma^2_g = 2 / f''(g)$, share $= 1 - \sigma^2_g / \sigma^2_{prior}$.
- **Finding:** with free zone factors, a single screenline count determines only **half** of the level, because the count can also be met by inflating the zones it crosses. Anchoring the level takes **several counts crossing independent zones**.

Synthetic benchmark: a city whose true OD is known



- Mean GEH on uncounted links falls from the prior's **5.6 to 2.2** (routes known) and **3.8** (routes also estimated). Freezing routes at the prior gives 8.0, so bilevel estimation is necessary.
- **96 % of true values lie within 2.5 standard errors:** the uncertainty is not over-confident.
- **Flows are identifiable; individual OD cells are not** (36 counts, 90 pairs). Reporting this openly is part of the contribution.

2 · Real data

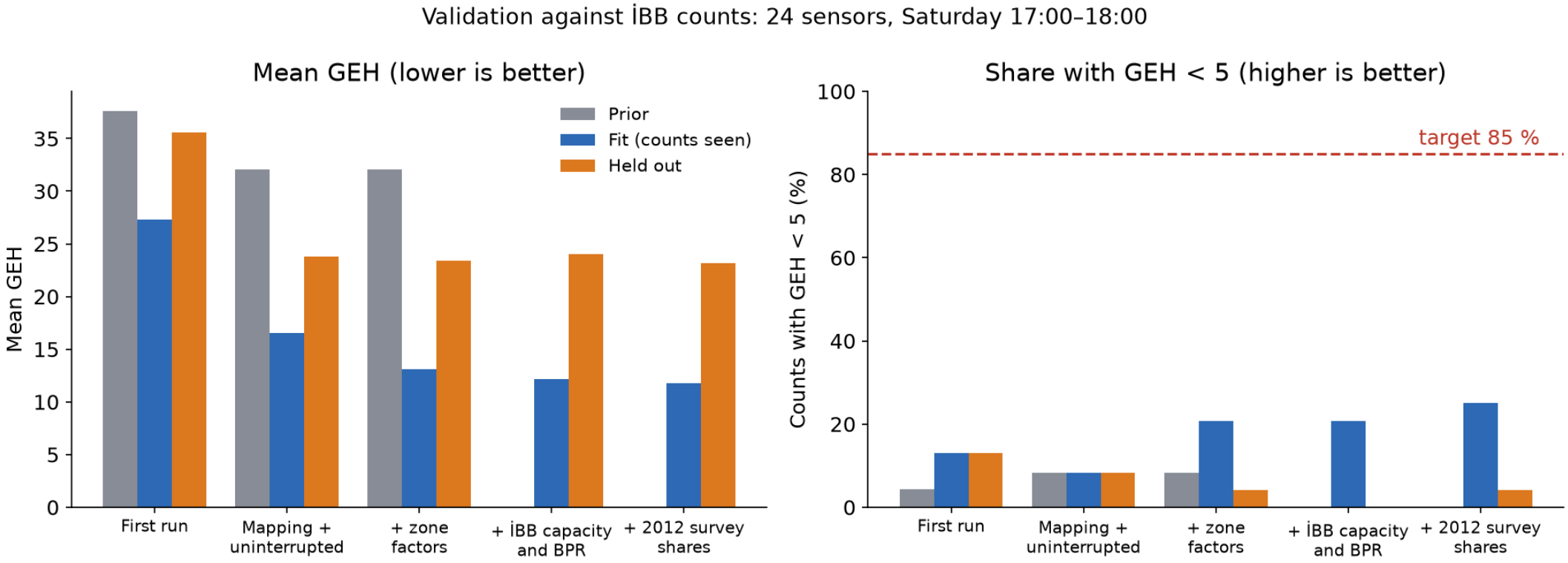
Count pipeline and mapping audit

- 25 of İBB's 27 sensors were mapped to **screenlines** across the road on the SUMO network (both directions, all lanes).
- **Daily to hourly:** using the hourly share of probe vehicles in the same geohash cell on the same day; 3.6–5.1 % of the day for Saturday 17:00–18:00.

Sensor	Daily	First mapping	Correction
Mecidiyeköy	110 k	side street (no route)	O-1
Çağlayan Junction	103 k	single-lane street	O-1
TEM Karanfilköy	103 k	Uğur Mumcu Street	O-2
FSM toll plaza	81 k	same edges as the bridge entrance	removed (toll lanes only)
TEM before FSM toll	98 k	both directions	one direction

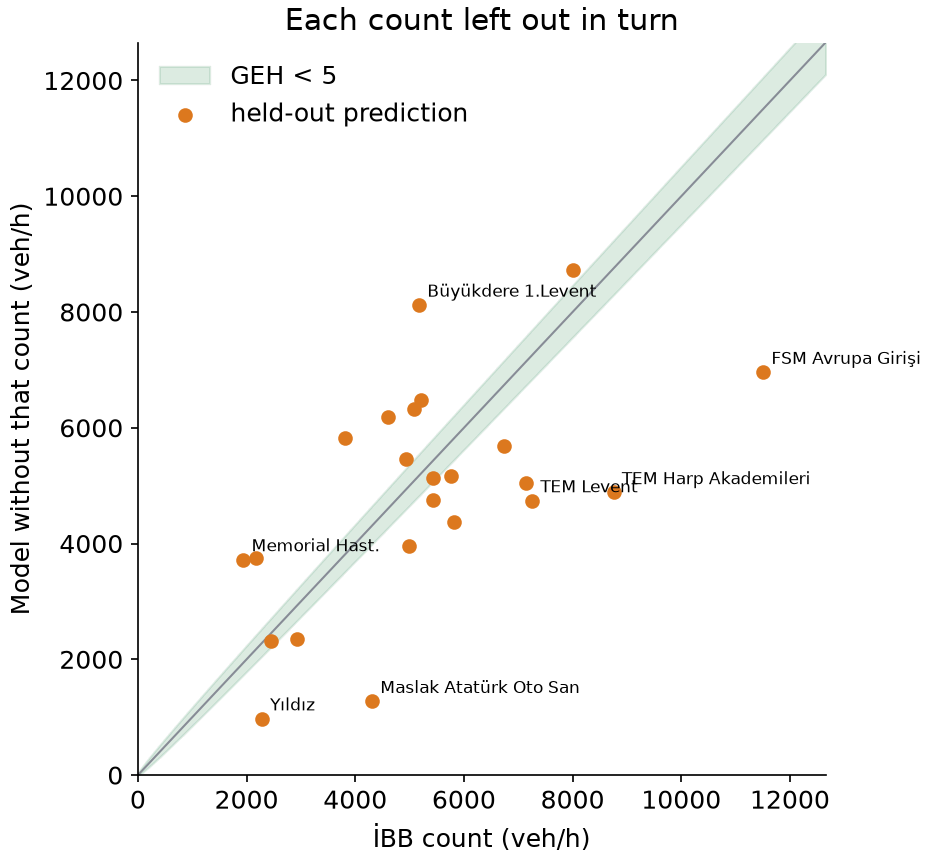
> On signalised arterials a low speed also comes from the red phase: lower bounds there inflated demand by 33 %. The rule is now restricted to **uninterrupted-flow** roads (motorway, trunk).

GEH results: progress, but the criterion is not met



Held-out mean GEH fell from 35.6 to 23.2, but the GEH < 5 criterion (85 %) is still far away.

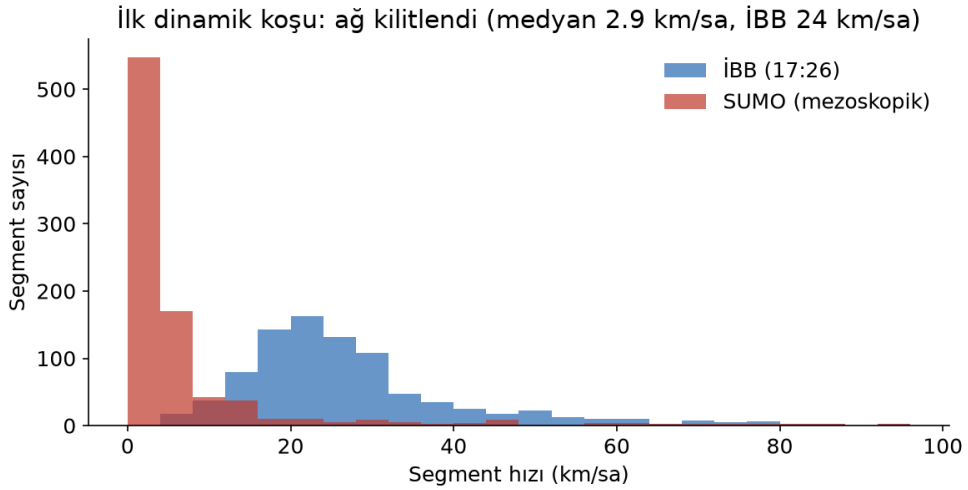
Where does the remaining error come from?



Monday 11:00 run: each count left out in turn.

1. **Count definitions conflict.** Three O-2 sensors settle on the same modelled flow, yet count 4,291 / 7,108 / 8,171 veh/h. Without ramps between them this is impossible: their direction or lane coverage must differ.
2. **Gateways and network.** At FSM the model can carry only half the count; the gateway prior is a crude capacity-based assumption.
3. **Prior.** Link parameters (İBB 2020 capacity and BPR) are not the bottleneck; trip-type shares from the 2012 survey bring a small improvement.

Dynamic test: SUMO

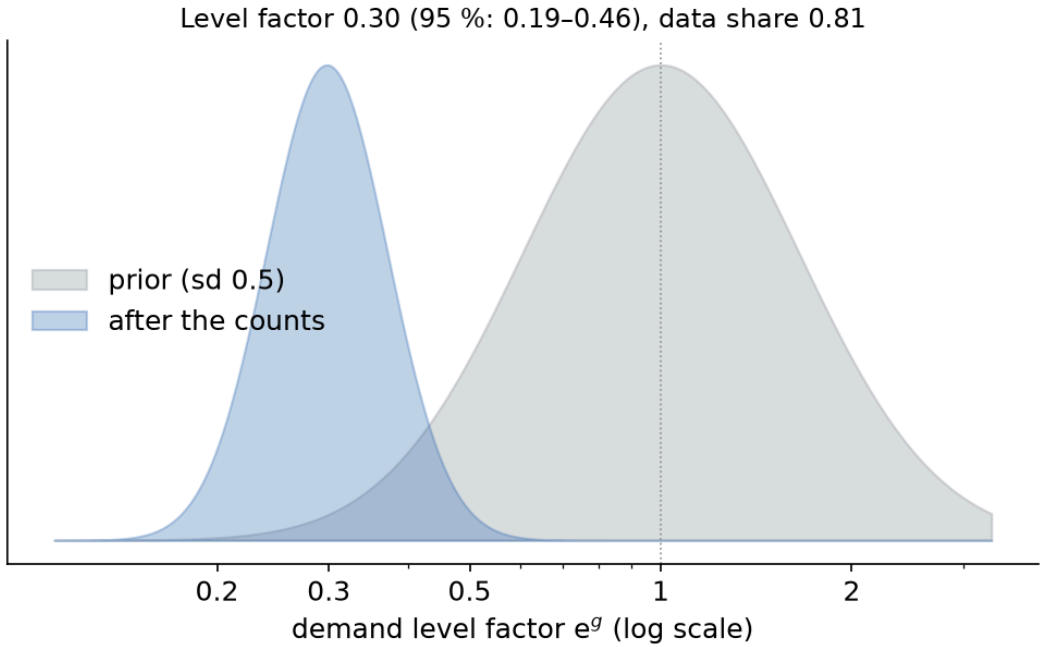


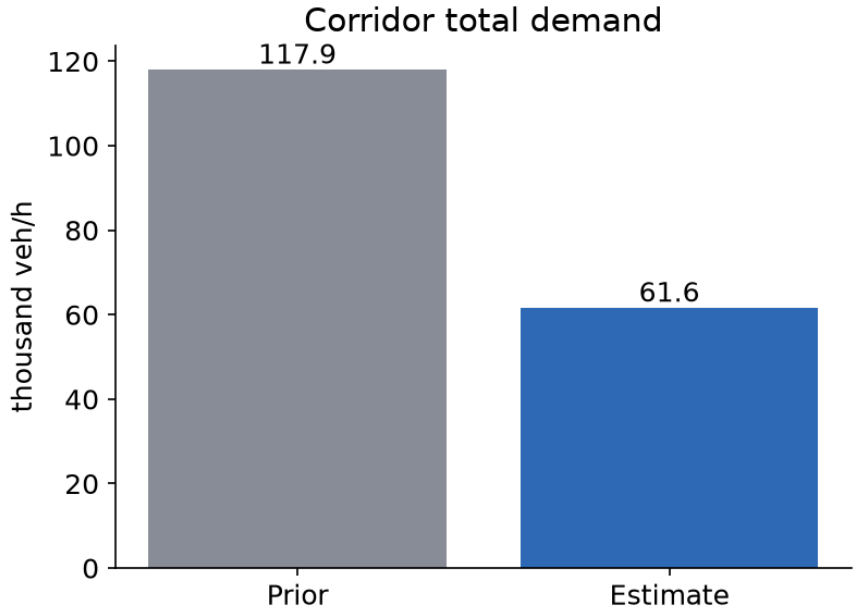
Segment speeds: İBB (blue) vs. SUMO mesoscopic run (red).

- Estimate without counts: 128.6 k veh/h, 90 min, mesoscopic model, dynamic rerouting.
- **The network gridlocked:** 102,735 of 192,551 vehicles could enter; median speed 2.9 km/h (İBB: 24 km/h).
- This confirms that demand not anchored by counts was far too high: the bridge count is 8.5 k veh/h, the estimate loaded 13.9 k. The estimate with counts is 74 k veh/h.

> This result directly motivates measuring the level of demand separately (Contribution 3).

Demand level on the real corridor: the counts anchor it





- Live İBB speeds on Monday at 11:30 and October 2024 Monday counts for 11:00–12:00 (23 sensors).
- Data share of the level **0.81**: for the first time the level is **anchored by counts** (threshold 0.8).
- Level factor **0.30** (95 %: 0.19–0.46); with the zone factors, total demand falls from 117.9 k to **61.6 k**: the land-use prior overstated demand **almost twofold**. This is the source of the gridlock in SUMO.
- Held-out share with GEH < 5 is 8.7 % (mean 22.9): **the level is right, the distribution is still weak.**

Two new modules: multi-source anchors and the neural auditor

Multi-source anchors

- An anchor constrains **a sum over the OD matrix**, not a link: one direction of the bridge, the trips a group of zones produces.
- **İstanbulkart:** 11,764 boardings between 11:00 and 12:00 at 31 rail and Metrobüs stops in the corridor.
- The counts had cut the corridor zones' production from 57.4 k to **15.6 k**; the independent İstanbulkart anchor (11.8 ± 11.9 k) **agrees**.

Simulation-to-reality network

- An MLP trained on 300 simulated samples (OD → speeds on 712 links); the MAP estimator is its **auditor**.
- R^2 0.15 for total demand. On live data it gave **4.8 times** the MAP estimate.
- Why: the static model does not produce real slowdowns (median speed ratio 0.32 in reality, 0.99 in the model).
- **The auditor caught it** (`trustworthy: false`). Next: samples generated by SUMO.

Shadow simulation: forking the future at an incident

- At the incident SUMO saves its whole state; from it every policy runs **faster than real time**. The simulation decides.
- Test: a five-lane stretch of the TEM closes at minute 20, simulated 20 minutes ahead (four runs in ~9 minutes).

Policy	Vehicles arrived	Teleports	Mean time loss
Road open (reference)	10,233	52	274 s
Unaware drivers	9,655	123	277 s
Informed at once	**10,390**	**50**	**271 s**

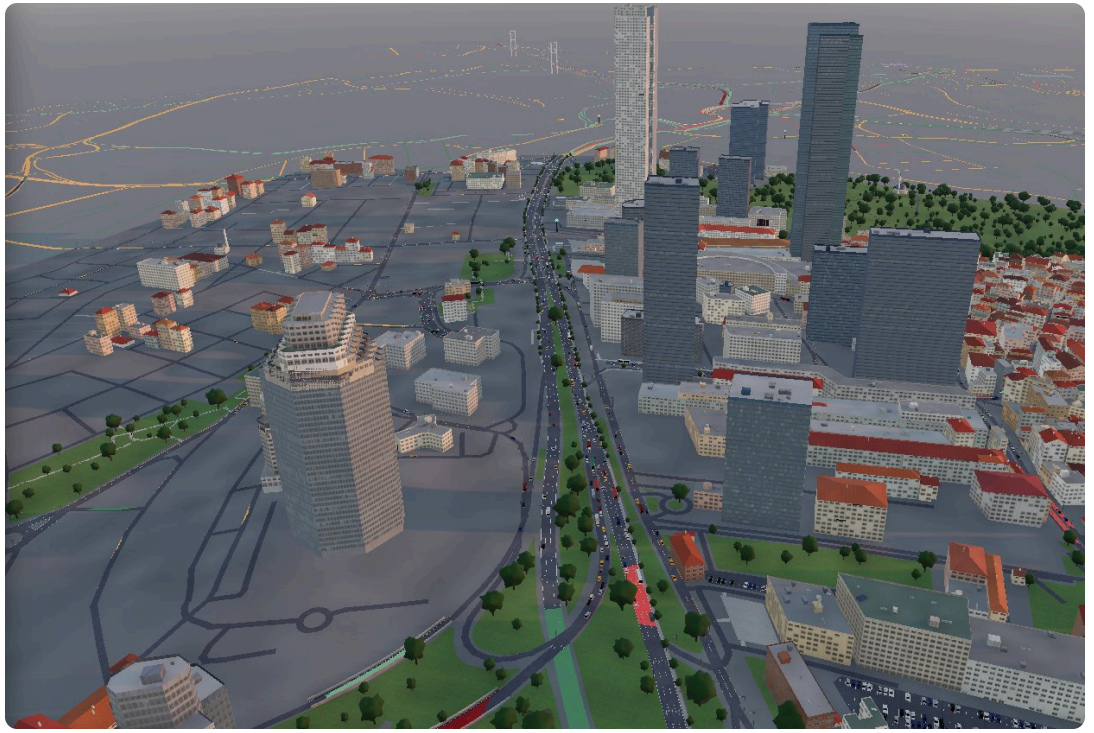
> **Verdict: inform at once** (+735 vehicles against unaware drivers). Honest note: this branch even beats the open road, because drivers also re-route more often; separating the two effects is next.

3 · The 3D digital twin

A procedural Istanbul from open data



15 July Martyrs Bridge (towers, main cables, hangers) and the Ortaköy Mosque



Levent: towers from OSM `building:part` data

Istanbul's urban fabric



Hipped tile roofs, chimneys, rooftop water tanks

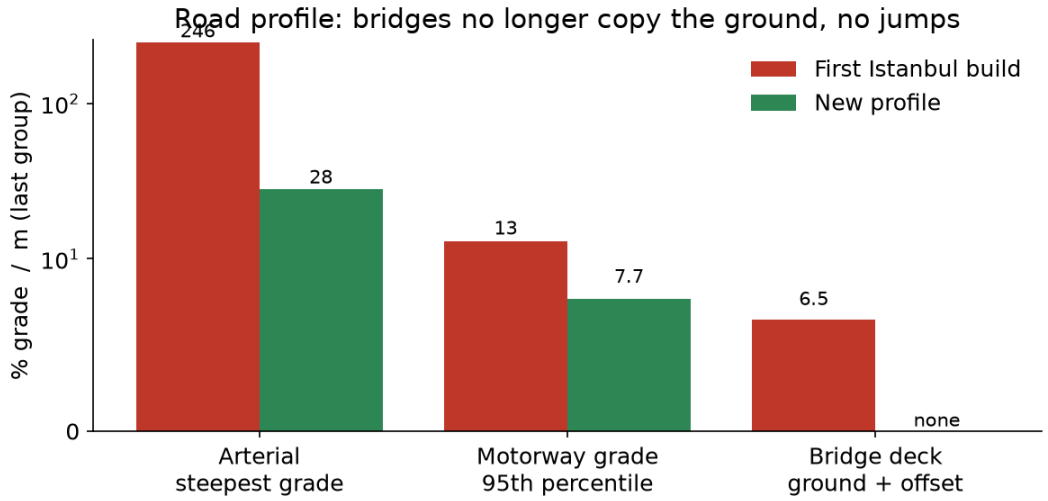


Shop fronts and signs with real shop names



Zincirlikuyu: multi-level interchange and viaducts

Road geometry



- **Bridges:** bridges over valleys or water get a flat deck from bank to bank.
- **Viaducts:** clearance above the ground below plus the layer; smoothed with vertical curves.
- **Tunnels** descend from the portal at 12 %; elevated roads are rounded with Chaikin's algorithm.
- **Performance:** 19.4 → 4.2 million triangles; 4,599 → 780 draw calls.

Limitations

- **The validation criterion is not met:** held-out share with GEH < 5 is 4.2 % (target 85 %).
- **The level is anchored, the distribution is not:** 23 counts anchor the level (data share 0.81), but a single bridge count cannot do so alone (0.53 in the synthetic test).
- **A single time slice:** one Saturday 17:00 snapshot; day-type and hourly variation are not yet modelled.
- **Static assignment:** no turn restrictions or queue spillback.
- **Hand-picked assumptions:** every prior parameter is listed in `summary.json`.
- **Literature:** the novelty of "speed as a censored observation" and "data share" must be confirmed by a systematic review.

Next steps

1. **Sensor definitions:** confirm direction, lane coverage and toll/ramp distinctions with İBB (the conflicting O-2 sensors).
2. **Level of demand:** several independent counts (bridge directions separately, O-1 and O-2 screenlines); data share of the level ≥ 0.8.
3. **Time:** day-type and hourly profiles from the hourly density archive; continuous live speed archiving.
4. **Dynamic consistency:** test count-anchored demand in SUMO with scaled runs (25 / 50 / 75 / 100 %).
5. **Recommendation engine:** departure time, mode and bridge choice for drivers; bottleneck ranking and scenario comparison for the authority.

Target publication path: TRB Annual Meeting / hEART (method and case), then *Transportation Research Part C* (multi-week validation) and an open dataset paper.

Reproducibility

```
npm run sumo:net -- --area=koridor          # OSM → SUMO network
npm run sumo:map-ibb -- --area=koridor      # İBB segments → edges
uv run --group sumo python tools/od/counts.py --area=koridor --month=2024-10 --weekday=5 --hour=17
uv run --group sumo python tools/od/corridor.py --area=koridor \
    --speeds data/od/koridor/20261017-1726/speeds.csv \
    --counts data/od/koridor/counts_202410_d5_h17.csv --tag counts
npm run test:od                             # synthetic validation (32 checks)
uv run --with matplotlib python tools/report/figures.py   # this deck's figures (TR and EN)
npm run dev                                 # 3D client: http://localhost:5280
```

Every assumption, intermediate result and negative finding is recorded with dates and numbers in the `tools/*/README.md` files.

References

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- UK Department for Transport. TAG Unit M3.1, Highway Assignment Modelling (GEH criterion).
- İBB Open Data Portal (data.ibb.gov.tr): Daily Vehicle Counts, Hourly Traffic Density, Transport Demand Forecasting Model Data.

Thank you

Questions and feedback

github.com/aaasiroglu/procedural-istanbul